

1. INTRODUCTION

Energy demand is increasing day by day, and there is a strong need to shift towards cleaner and more sustainable sources. Solar energy is one of the most promising options because it is renewable and widely available. As a result, solar panels are being used in homes, industries, and large-scale power plants.

However, one of the practical challenges in using solar panels is maintaining their efficiency over time. Environmental conditions such as dust, pollution, and weather changes can affect their performance. In regions where dust levels are high, this becomes an even bigger concern, as it directly impacts energy generation.

Because of these challenges, improving the maintenance of solar panels has become an important area of focus. Finding simple and effective ways to keep panels functioning efficiently can make solar energy systems more reliable and useful in the long run.

1.1 Existing systems & their limitations

1.1.1 Overview of Existing Systems

Right now, there are a few common ways people keep solar panels clean. These methods do the job to some extent, but most of them follow a fixed pattern and don't really consider what's actually happening around the panels, like how much dust has settled or what the weather is like.

[1] Manual Cleaning Methods

This is the most basic approach. Panels are cleaned by hand using water, cloth, or brushes. It's mostly used in homes or small setups where there aren't too many panels. While it's simple, it depends completely on someone remembering to clean the panels regularly, which doesn't always happen.

[2] Water-based cleaning system

In this method, water is sprayed over the panels using sprinklers or pipes to wash away dust. It's more useful in bigger setups where manual cleaning is difficult. But the system usually works at fixed times, whether the panels actually need cleaning or not.

[3] Robotic Cleaning Systems

These systems use machines that move over the panels and clean them using brushes. They are mostly used in large solar farms. They reduce human effort, but they also need proper setup, maintenance, and can be quite costly.

1.1.2 Limitations and Drawbacks of Existing Systems

Even though these systems are useful in maintaining solar panels, they have several drawbacks that reduce their effectiveness in real-life situations:

[1] Dependence on Manual Effort: Manual cleaning requires people to clean the panels regularly, which can be tiring and time-consuming. In large installations, this becomes difficult to manage and may lead to irregular cleaning.

[2] High Water Consumption: Water-based systems use a large amount of water for cleaning. This can be a serious issue in areas where water is scarce. Also, frequent use of water increases operational cost over time.

[3] High Cost of Advanced Systems: Robotic cleaning systems are expensive to install and maintain. Because of this, they are not suitable for small-scale users or low-budget projects.

[4] Lack of Smart Decision-Making: Most existing systems work on fixed schedules and do not consider real-time conditions. They cannot differentiate between dust accumulation and temporary changes like cloudy weather, which can lead to unnecessary cleaning or missed cleaning.

[5] Limited Adaptability: These systems do not adjust based on environmental changes such as weather conditions, dust levels, or temperature variations. As a result, their performance is not always efficient in different situations.

1.2 Problem statement

Solar panels often lose efficiency over time due to the accumulation of dust and changing environmental conditions. Cleaning them manually can be difficult, time-consuming, and not always practical. Many existing systems also lack smart decision-making. Therefore, there is a need for an automated system that can detect these conditions and clean the panels efficiently while reducing human effort.

1.3. Objectives

- [1] To develop an automatic solar panel cleaning system
- [2] To detect dust using a light sensor and to avoid unnecessary cleaning using temperature sensor
- [3] To control the motor using ESP32
- [4] To enable IoT monitoring using ThingSpeak .

2.0 LITERATURE SURVEY

[1] Mani, M. and Pillai, R. (2010) studied how dust settles on solar panels and reduces their efficiency. They showed that even a thin layer of dust can block sunlight and affect power generation. However, their work mainly focuses on analysis and does not suggest any automatic cleaning methods.

[2] Sayyah, A., Horenstein, M. and Mazumder, M. (2014) explored different cleaning methods such as brushing, water cleaning, and electrostatic techniques. They found that regular cleaning helps improve performance. But these methods are often expensive and not easy to implement for small or simple systems.

[3] Kumar, N. et al. (2018) developed a robotic system that uses a motor and brush to clean solar panels. This reduced manual work and made the process easier. However, the system works on a fixed schedule and cannot decide when cleaning is actually needed.

[4] Patil, S. and Deshmukh, P. (2019) designed a microcontroller-based cleaning system that operates using a timer. It cleans the panels automatically at set intervals. The drawback is that it may clean the panels even when they are already clean, leading to unnecessary energy usage.

[5] Rakesh, R. and Kumar, V. (2020) introduced a system using an LDR sensor to detect changes in light intensity. When the light decreases, the system assumes dust is present and starts cleaning. However, this method is not fully reliable because light can also decrease due to clouds or weather conditions.

[6] Sharma, A. and Gupta, R. (2021) proposed a solar-powered cleaning system using DC motors and brushes. Their design is energy-efficient and suitable for remote areas. However, it does not include any monitoring system or smart decision-making capability.

[7] Singh, P. and Kaur, H. (2022) developed an IoT-based monitoring system using ESP32 and ThingSpeak. This system allows users to monitor solar panel conditions remotely. However, it focuses only on monitoring and does not include automatic cleaning.

[8] Ahmed, S. and Rahman, F. (2022) worked on improving motor control systems for automation. Their study helped in achieving better motor performance. However, their work does not consider environmental sensing or cleaning requirements.

3.0 SOFTWARE AND HARWARE

3.1 Software description

3.1.1 Software Development and Tech Stack

The project was designed using the ESP32 microcontroller that was programmed in Embedded C using the Arduino IDE. The ESP32 was selected because it contains integrated Wi-Fi connectivity making it easier to incorporate IoT functions into the design. The Arduino IDE was selected for programming because of its user-friendly interface and adequate support libraries. The ESP32 can connect to the Wi-Fi network and set up a rudimentary web server where the user can control the cleaning system using their phone. The software deals with the functionality of controlling the motor and timing.

3.1.2 Integrated Development Environments (IDE)

The design and development of the embedded program was done using the Arduino IDE. It served as the tool where the code was developed, compiled, and uploaded to the ESP32.

Arduino IDE

The Arduino IDE was utilized in developing the embedded code for the ESP32. It is user-friendly and provides various library support needed for Wi-Fi communication and control of the motor. The embedded code was developed using Embedded C language.

The system does not have an external mobile app as its user interface. A web server runs on the ESP32 to control the system through any mobile web browser.

3.1.3 Programming Languages

The system has been built utilizing the programming language Embedded C/C++ with the help of the Arduino IDE. This language was applied in writing control algorithms for the ESP32 microcontroller.

Embedded C/C++ makes it possible to effectively interact with the hardware elements, such as a motor driver, and control time operations precisely. In addition, this language offers the opportunity to establish Wi-Fi communication.

3.1.4 UI/UX Framework

The interface of the system is a simple web interface which is available at the ESP32. The interface is built using HTML coding. This interface makes use of buttons to enable the users to start or stop the cleaning activity.

The web interface offers simplicity and faster access without the need of a mobile application.

3.1.5 Wireless Communication Implementation

Communication is enabled via Wi-Fi communication carried out using the ESP32 controller. In this case, the ESP32 controller connects to the Wi-Fi network and runs the web server.

Access to the system is gained using the mobile phone through typing the IP address of the ESP32 controller in the web browser. User commands are sent using HTTP request messages that are received by the ESP32 controller for motor control.

Thus, no additional communication modules are required for wireless communication.

3.2 Hardware description

3.2.1 Photovoltaic Solar Panel

This PV solar panel is utilized in the system as a demo component. This serves as the surface that needs cleaning to increase its performance. The dust collected over the surface of the solar panel reduces its capacity to capture light energy.

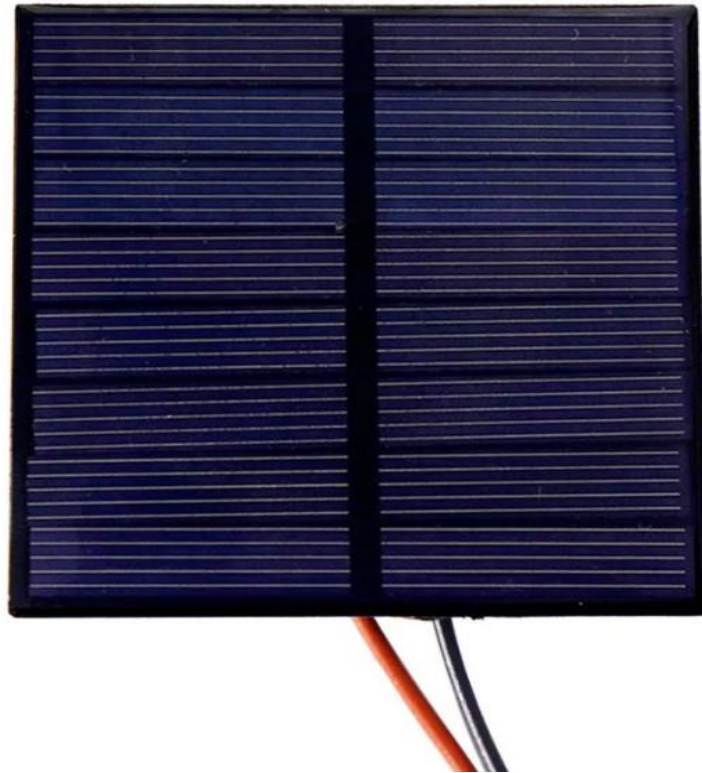


Fig 1: Solar Panel

For the purpose of this project, a mini solar panel will be used as an actual representation of the situation. The system involves cleaning of the panel using the rotating brush/wiper.

3.2.2. ESP32 Microcontroller

The ESP32 acts as the main control unit of the system. It processes user commands received through Wi-Fi and controls the motor accordingly. Its built-in Wi-Fi capability enables IoT-based operation.



Fig 2: ESP32 Microcontroller

As shown in Fig 2, the ESP32 development board serves as the central processing unit, interfacing with the sensors, relay module, and other system components.

3.2.3. L298 Motor Driver

The L298 motor driver is used to control the DC motor in the system.



Fig 3: L298 Motor Driver

It acts as an interface between the ESP32 and the motor, enabling safe and efficient motor operation. As shown in Fig 3, the motor driver receives control signals from the ESP32 and drives the motor accordingly.

3.2.4. DC Servo Motor

The DC servo motor is used as the cleaning mechanism of the system. It converts electrical energy into controlled mechanical motion to perform the cleaning operation. As shown in Figure X, the servo motor is connected to the motor driver and operates according to the control signals received from the ESP32.



Fig 3: DC Servo Motor

As shown in Fig 3, the servo motor is connected to the motor driver and operates according to the control signals received from the ESP32.

3.2.5. Power Supply

A DC power supply (adapter or battery) is used to power the system components, including the ESP32 and motor driver.

4.0 METHODOLOGY

4.1 System Architecture and Integration Overview

The proposed Internet of Things based solar panel cleaning system is designed to check the condition of a panel and clean it when dust builds up and affects its performance. This solar panel cleaning system uses a few parts like sensors and a small computer to make decisions and communicate with the internet.

First a special sensor called a Light Dependent Resistor checks how much light is hitting the panel. When dust gets on the panel less light gets through. The sensor readings change. But sometimes the light can be blocked by clouds so we also use a temperature sensor to get an idea of what is going on outside.

The sensor readings are. Looked at by a small computer called the ESP32, which is like the brain of the solar panel cleaning system. The ESP32 looks at the sensor data. Decides if the problem is dust on the panel. If it thinks the dust is the problem, it starts the cleaning process.

The ESP32 is connected to the internet. Sends the sensor data to a website called Thing Speak, where it is stored and shown as graphs. This allows people to check on the panel from far away and see how well the solar panel cleaning system is working overtime.

When the solar panel needs to be cleaned the ESP32 sends a signal to a motor that makes the cleaning brush move back and forth across the panel. The brush wipes away the dust. Then stops when it is done.

Here is how the solar panel cleaning system works:

- * The Light Dependent Resistor and temperature sensors check the environment.
- * The ESP32 looks at the sensor readings.
- * The sensor data is sent to Thing Speak over the internet.

- * Thing Speak stores the data. Shows it as graphs.
- * The ESP32 decides if the solar panel needs to be cleaned.
- * The motor makes the cleaning brush move.
- * The cleaning brush wipes the dust off the panel.
- * The motor stops when it is done.

This way the solar panel cleaning system can check on the panel show the data, on the internet and clean the panel by itself which makes it more efficient and saves people from having to do it manually.

4.2 Hardware Integration and Physical Execution

The hardware components are interconnected to perform the cleaning operation efficiently.

[1] Control Unit:

The ESP32 receives commands through Wi-Fi and processes them to control the system.

[2] Motor Driver Interface:

The L298 motor driver acts as an interface between the ESP32 and the DC motor, allowing safe and controlled motor operation.

[3] Cleaning Mechanism:

The DC motor is attached to a brush or wiper that physically cleans the solar panel surface when activated.

4] Power Supply:

The system is powered using a DC supply (adapter or battery), ensuring proper operation of all components. The cleaning process is initiated through the web interface, where the user sends a command to start the motor. The motor runs for a predefined duration and then stops

automatically.

4.3 Software Architecture and Supervisory Control

This program is developed in the Arduino IDE and is burned into the ESP32 microcontroller. This system incorporates Wi-Fi capabilities, web server configuration, and motor control code. Once the ESP32 microcontroller connects to the Wi-Fi network, it gets an IP address. A user can then log in to that IP address using their web browser to gain access to the control panel. The commands entered in the control panel are processed by the ESP32 microcontroller to control the motor. In addition, there is a timer function that shuts down the motor after a specific time interval.

5.0 RESULTS AND DISCUSSIONS

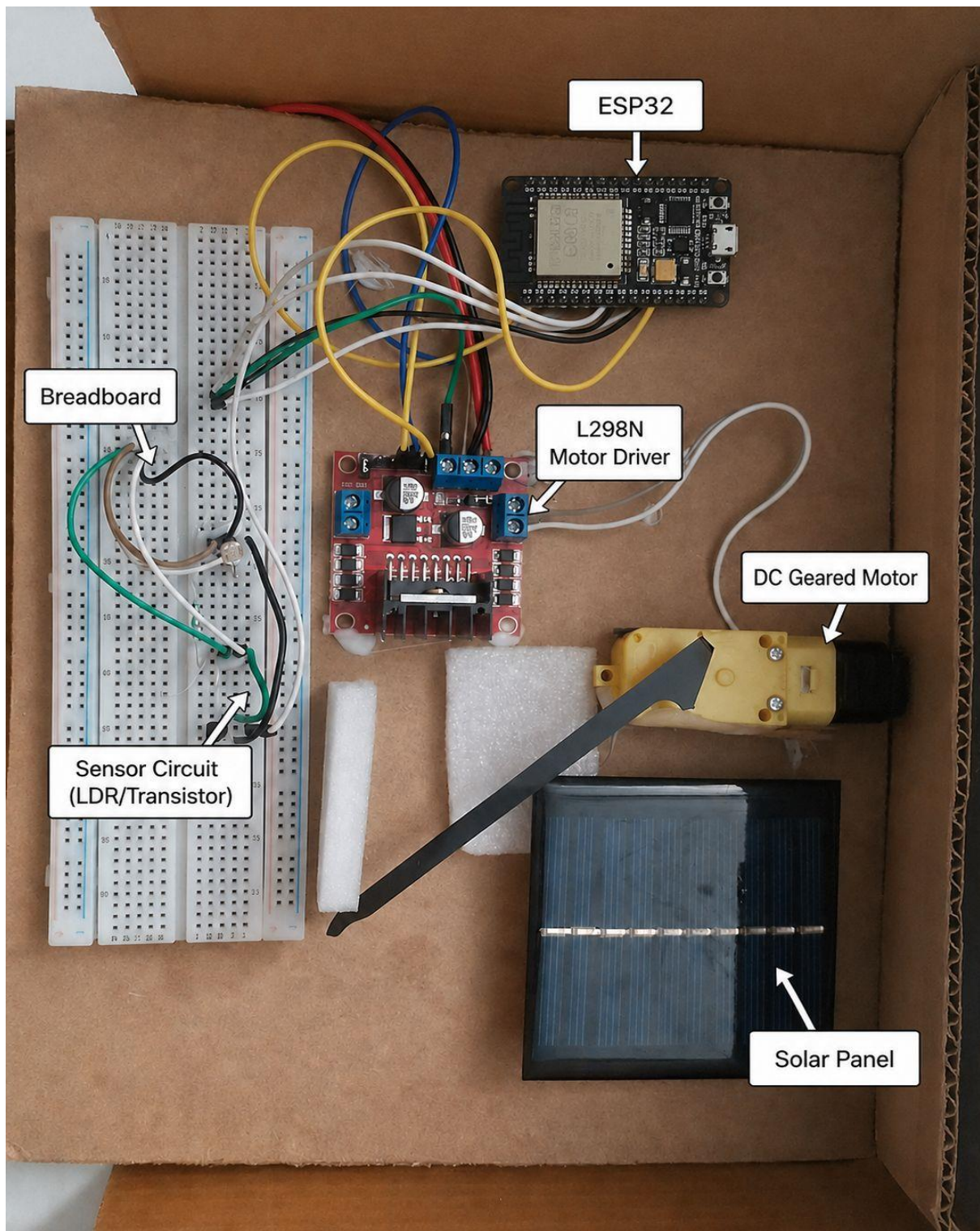


Fig 4: Hardware setup of the solar panel cleaning system

The ESP32 sends sensor data to the ThingSpeak cloud platform using Wi-Fi. This data was. Shown on the ThingSpeak dashboard using charts.

We could see what was happening with the temperature sensors in real time on these charts. The ThingSpeak cloud platform made it easy to watch what was going on with the system from away without needing to be right next to the hardware.

The ThingSpeak platform worked well. Helped us see what was happening with the sensors. The ThingSpeak platform made it easier to keep an eye on the system and let users look at sensor data from times.

The ThingSpeak platform and the ESP32 worked together to send sensor data to the cloud. This made it easy to use the ThingSpeak platform to monitor the system and look at the sensor data.

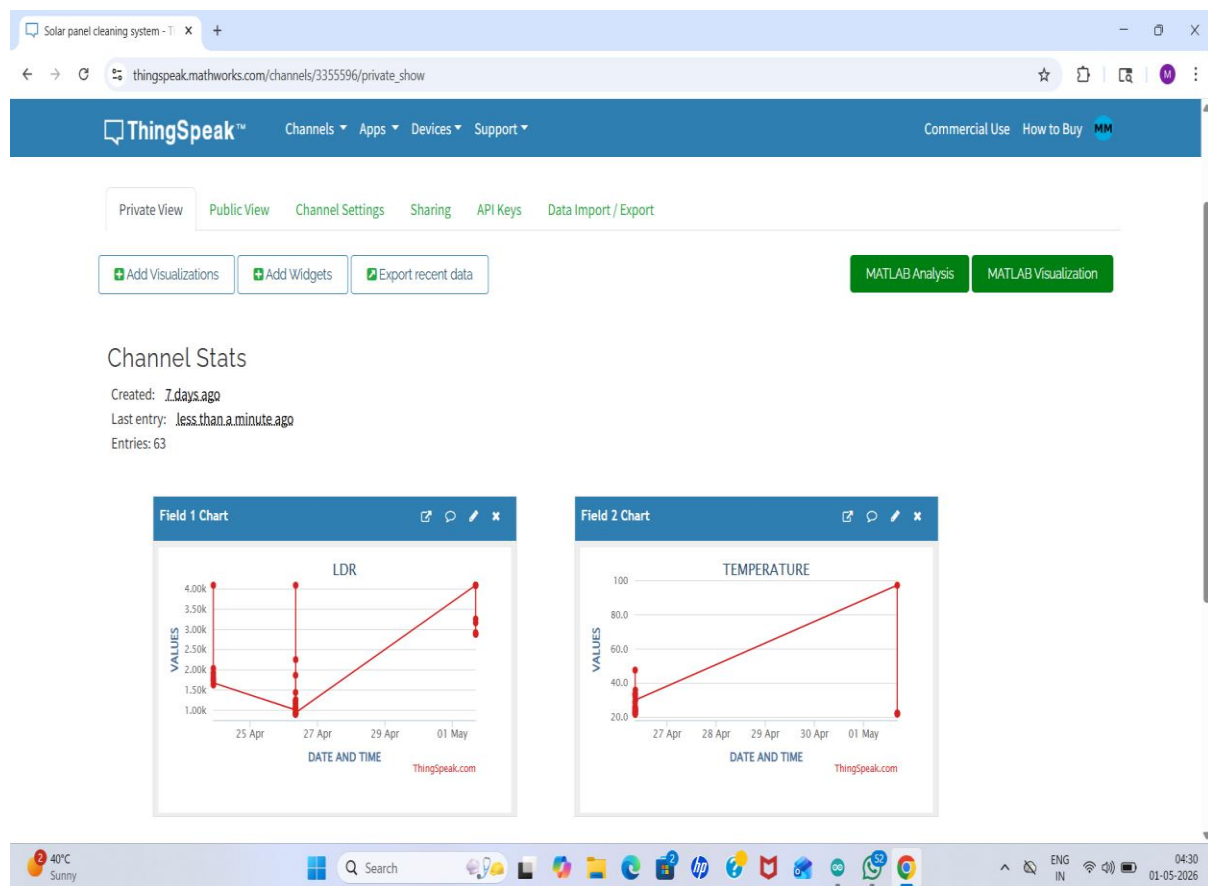


Fig 5: ThingSpeak Graph of Temperature Sensor Readings and LDR Sensor Readings

The ThingSpeak platform is a tool, for monitoring sensor data. The ESP32 and the ThingSpeak platform worked well together to send and show sensor data.

5.1 Prototype Realization and System Integration

The prototype for the IoT-enabled Automated Solar Panel Cleaning System was successfully constructed with the help of the ESP32 microcontroller, motor driver, DC motor, and solar panel module.

The prototype was evaluated to confirm the appropriate communications between the ESP32 and the motor driver via the Wi-Fi control method. Motor performance was successfully carried out in response to user instructions from the web interface.

5.2 System Performance and Response Time

The system was put to the test to analyze the responsiveness of the system. The ESP32 was able to connect to the Wi-Fi network and host a web interface.

Upon entering the command from the mobile browser, there was no noticeable delay in the reaction of the motor. The system performed efficiently and reliably throughout all tests conducted.

5.3 Cleaning Efficiency

Cleaning was carried out using the proposed mechanism on a dusty surface of the solar panel. The brush driven by the motor was able to clean the surface efficiently.

5.4 Overall System Performance

[1] The performance of the system was assessed using factors such as reliability, usability, and effectiveness.

[2] The system was reliable, with no serious breakdowns occurring during testing

[3] The Web interface was easy to use and navigate

[4] The cleaning was efficient

[5] The system responded quickly

The above results prove that the suggested Internet of Things solar panel cleaning system is efficient and appropriate for small scale applications.

5.5 Power Usage and Efficiency

This system was intended to have minimal energy consumption. The ESP32 and motor driver use very little power, making this system very efficient in its energy use for small projects.

DC motors will be in use only when there is a cleaning operation; hence unnecessary energy consumption is eliminated. Because the system will work for a short while during each cleaning session, the energy consumption will not be high.

One can choose a solar panel to act as a secondary power source.

5.6 Comparative Analysis with Conventional Methods

The proposed system was compared with traditional manual cleaning methods.

Manual Cleaning:

Requires human effort, is time-consuming, and may not be practical for large installations.

Proposed System:

Provides automated cleaning using a motor-driven mechanism, reducing human effort and improving efficiency.

The system offers advantages such as reduced labour, consistent cleaning, and ease of operation through IoT-based control. This makes it a practical solution for maintaining solar panel efficiency.

FUTURE SCOPE

There are several areas where the proposed IoT-based Automatic Solar Panel Cleaning System may be enhanced. Additional sensors like dust sensors and rain sensors may be introduced in the system to allow automatic operation of the cleaning process without any human intervention.

This system can be further enhanced by adding an extra component like a battery in combination with a solar power unit which will make the system entirely independent. Automation techniques like rail automation or robotic movement can also be introduced for large scale operations.

REFERENCES

- [1] M. Mani and R. Pillai, "Impact of dust on solar photovoltaic (PV) performance: Research status, challenges and recommendations," *Renewable and Sustainable Energy Reviews*, vol. 14, no. 9, pp. 3124–3131, 2010.
- [2] A. Sayyah, M. N. Horenstein, and M. K. Mazumder, "Energy yield loss caused by dust deposition on photovoltaic panels," *Solar Energy*, vol. 107, pp. 576–604, 2014.
- [3] N. Kumar, R. Singh, and A. Sharma, "Design and development of a robotic solar panel cleaning system," *International Journal of Engineering Research and Technology (IJERT)*, vol. 7, no. 5, pp. 45–49, 2018.
- [4] S. Patil and P. Deshmukh, "Microcontroller-based automatic solar panel cleaning system," *International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering (IJIREEICE)*, vol. 7, no. 4, pp. 120–124, 2019.
- [5] R. Rakesh and V. Kumar, "LDR-based intelligent solar panel cleaning mechanism," *International Journal of Scientific Research in Engineering and Management (IJSREM)*, vol. 4, no. 8, pp. 1–5, 2020.
- [6] A. Sharma and R. Gupta, "Solar-powered automated cleaning system for photovoltaic panels," *International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)*, vol. 1, no. 5, pp. 78–83, 2021.

[7] P. Singh and H. Kaur, "IoT-based monitoring system for solar panels using ESP32 and Thing Speak," *International Journal of Engineering Research and Applications (IJERA)*, vol. 12, no. 3, pp. 34–39, 2022.